A tensor product formula for Efimov rigidification on p-adic formal schemes

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Abstract

Let $\mathfrak{S} = \operatorname{Spf} \mathbb{Z}_p$, and let $\mathfrak{X}$ and $\mathfrak{Y}$ be quasi-compact, quasi-separated p-adic formal schemes locally topologically of finite type over $\mathfrak{S}$. We prove that exterior product induces an isomorphism

$$\mathrm{QC}(\mathfrak{X})^{\mathrm{rig}} \otimes_{\mathrm{QC}(\mathfrak{S})^{\mathrm{rig}}} \mathrm{QC}(\mathfrak{Y})^{\mathrm{rig}} \simeq \mathrm{QC}(\mathfrak{X} \times^{\mathrm{der}}_{\mathfrak{S}} \mathfrak{Y})^{\mathrm{rig}},$$

where rig denotes Efimov rigidification and the product on the right is the derived formal fiber product. If either factor is flat over $\mathbb{Z}_p$, this is the ordinary formal fiber product. In particular, the theorem applies to p-adic completions of finite type $\mathbb{Z}$-schemes. The affine proof combines a Beck–Chevalley criterion with coherent generation of the generic diagonal and a separate argument for p-torsion. Zariski descent then gives the statement for formal schemes. We also give a counterexample to the ordinary-product formulation without a flatness hypothesis.

1 Statement and conventions

Fix a prime p, and put

$$R = \mathbb{Z}_p, \qquad K = \mathbb{Q}_p, \qquad \mathfrak{S} = \operatorname{Spf} R.$$

Throughout, categories are ∞-categories, modules are module spectra, and tensor products of rings are tensor products of commutative algebras in spectra. An ordinary ring is identified with its Eilenberg–Mac Lane algebra. Thus no flatness is implicit in the notation $A \otimes_R B$. The symbol Hom denotes a mapping spectrum; an internal mapping object is denoted by $\underline{\mathrm{Hom}}$. A continuous functor is a functor preserving all small colimits. All categorical tensor products are formed in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$, or in the indicated category of modules in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$.

For a commutative R-algebra T, write

$$M^{\wedge}_p = \lim_{n \geq 1} (M \otimes_R R/p^n), \qquad \mathcal{C}(T) = \mathrm{Mod}_T^{\wedge_p}.$$

Here $\mathcal{C}(T)$ is the category of p-complete T-modules. Its tensor product is completed tensor product, its unit is $T^{\wedge}_p$, and its colimits are completed module colimits. We use only p-complete algebras T, so the unit is T. We write

$$M \mathbin{\widehat{\otimes}}_T N = (M \otimes_T N)^{\wedge}_p, \qquad A \mathbin{\widehat{\otimes}}_R B = (A \otimes_R B)^{\wedge}_p.$$

When the subscript is K, the notation $\widehat{\otimes}_K$ instead denotes the ordinary nonarchimedean completed tensor product of affinoid algebras or finite affinoid modules. For an affine formal scheme, the convention for quasi-coherent modules is

$$\mathrm{QC}(\operatorname{Spf} T) = \mathcal{C}(T).$$

On a formal scheme these categories are glued by Zariski descent. Put

$$\mathcal{N}(T) = \mathcal{C}(T)^{\mathrm{rig}}, \qquad \mathcal{N}(\mathfrak{X}) = \mathrm{QC}(\mathfrak{X})^{\mathrm{rig}}, \qquad \mathcal{E} = \mathcal{N}(R).$$

The rigidification is the one studied by Efimov [3, §§1.5,4]. In particular, the superscript rig in this note is a categorical construction.

Definition 1.1. The formal schemes considered in the theorem are quasi-compact and quasi-separated, with affine open charts of the form

$$\operatorname{Spf} A, \qquad A = R\langle t_1, \dots, t_d\rangle / I, \qquad R\langle t_1, \dots, t_d\rangle = \lim_n (R/p^n)[t_1, \dots, t_d],$$

equipped with the p-adic topology. Here I is an ideal; it is automatically finitely generated and closed. Equivalently, the structure morphism to $\mathfrak{S}$ is adic and locally topologically of finite type [9, Tag 0GBW]. No flatness condition is part of this definition.

For such $\mathfrak{X}$ and $\mathfrak{Y}$, their derived formal fiber product is obtained on pairs of affine charts by

$$\operatorname{Spf} A \times^{\mathrm{der}}_{\mathfrak{S}} \operatorname{Spf} B = \operatorname{Spf}(A \mathbin{\widehat{\otimes}}_R B). \tag{1.1}$$

This may have a non-discrete structure sheaf. Its ordinary truncation is the ordinary formal fiber product.

Theorem 1.2. *Let $\mathfrak{X}$ and $\mathfrak{Y}$ satisfy Definition 1.1. Exterior product induces a canonical isomorphism of presentable stable symmetric monoidal categories*

$$\mathcal{N}(\mathfrak{X}) \otimes_{\mathcal{E}} \mathcal{N}(\mathfrak{Y}) \xrightarrow{\sim} \mathcal{N}(\mathfrak{X} \times^{\mathrm{der}}_{\mathfrak{S}} \mathfrak{Y}). \tag{1.2}$$

If either $\mathfrak{X}$ or $\mathfrak{Y}$ is flat over R, the derived formal fiber product in (1.2) is the ordinary formal fiber product.

For affine charts this states

$$\mathcal{N}(A) \otimes_{\mathcal{E}} \mathcal{N}(B) \simeq \mathcal{N}(A \mathbin{\widehat{\otimes}}_R B). \tag{1.3}$$

The comparison sends an exterior pair (P, Q) to the tensor product of its pullbacks to the formal product. The affine proof is given in Section 5, and the passage to formal schemes in Section 6.

2 Complete modules and change of scalars

We first record the categorical input, including the change-of-scalars statement needed in the proof.

2.1 The rings and smallness conditions

Call a p-complete connective commutative R-algebra T *bounded noetherian* if $\pi_0 T$ is noetherian, each $\pi_i T$ is finite over $\pi_0 T$, and $\pi_i T = 0$ for all sufficiently large i. For such T, let $\mathrm{Coh}^b(T) \subset \mathrm{Mod}_T$ be the full stable category of modules with finitely many nonzero homotopy groups, each finite over $\pi_0 T$. Let $\mathrm{Perf}(T)$ denote the category of perfect T-modules. For a collection of objects in a stable category, thick denotes the smallest full stable subcategory closed under retracts and containing that collection. In a presentable stable category, Loc denotes closure under all colimits and shifts.

Lemma 2.1. *Let A and B be ordinary topologically finite type R-algebras, complete for the p-adic topology. Then $C = A \mathbin{\widehat{\otimes}}_R B$ is bounded noetherian, $\pi_0 C$ is topologically of finite type over R, and*

$$C[1/p] \simeq A[1/p] \mathbin{\widehat{\otimes}}_K B[1/p]. \tag{2.1}$$

The right side is the ordinary completed tensor product of affinoid K-algebras; in particular, it is discrete. If P and Q are finite discrete modules over A and B, respectively, then $P \mathbin{\widehat{\otimes}}_R Q$ is bounded coherent over C, and

$$(P \mathbin{\widehat{\otimes}}_R Q)[1/p] \simeq P[1/p] \mathbin{\widehat{\otimes}}_K Q[1/p].$$

Proof. Choose presentations of A and B as quotients of $S = R\langle x_1, \dots, x_a\rangle$ and $T = R\langle y_1, \dots, y_b\rangle$. These are regular noetherian rings of finite dimension. A finite module over either ring is therefore perfect. Put

$$H = S \mathbin{\widehat{\otimes}}_R T = R\langle x_1, \dots, x_a, y_1, \dots, y_b\rangle.$$

Associativity identifies the completed exterior product of finite S- and T-modules P, Q with

$$(P \otimes_S H) \otimes_H (H \otimes_T Q).$$

The displayed object is already complete, since its factors are perfect over the complete noetherian ring H. Thus completed exterior products can be calculated by extending finite perfect presentations from S and T to H and tensoring there. This gives bounded homotopy groups finite over H. For C, these groups are modules over

$$\pi_0 C = H/(I_A, I_B),$$

so they are finite over $\pi_0 C$. The same argument applies to $P \mathbin{\widehat{\otimes}}_R Q$.

After inverting p, the same presentations calculate completed exterior products over K. Completed tensor product over a complete nonarchimedean field is exact separately on finite affinoid modules: a short exact sequence of such modules is a strict sequence of Banach spaces of countable type and splits continuously as a sequence of K-Banach spaces. Tensoring the splitting proves exactness. See [2, proof of Lemma 1.1.5] and [1, §§2.7.1,3.7.3]. This proves (2.1) and its discreteness. □

For a stable category $\mathcal{C}$, write $\mathcal{C}^{\omega_1}$ for its full subcategory of ω_1-compact objects, where ω_1 is the first uncountable cardinal.

Lemma 2.2. *Let $S \to T$ be a map of bounded noetherian p-complete commutative R-algebras, and suppose that $\pi_0 T/p$ is of finite type over $\pi_0 S/p$. The adjunction*

$$f : \mathcal{C}(S) \rightleftarrows \mathcal{C}(T) : u, \qquad f(M) = T \mathbin{\widehat{\otimes}}_S M,$$

has the following properties:

(i) *Both categories are locally rigid, their units are ω_1-compact, and their ω_1-compact objects are closed under tensor product.*

(ii) *Both f and u preserve colimits and ω_1-compact objects.*

(iii) *The completed projection formula holds:*

$$u\big(f(M) \mathbin{\widehat{\otimes}}_T N\big) \simeq M \mathbin{\widehat{\otimes}}_S u(N).$$

Proof. The category $\mathcal{C}(S)$ is compactly generated by

$$K_S = \mathrm{cofib}(p : S \to S) = S \otimes_R \mathbb{F}_p.$$

Indeed, mapping out of K_S detects reduction modulo p and commutes with completed colimits; reduction modulo p is conservative on complete modules. The object K_S is dualizable, since it is perfect as an S-module. Thus all compact objects of $\mathcal{C}(S)$ are dualizable, tensor products of compact objects are compact, and the multiplication has a continuous bimodule right adjoint. Since a compactly generated presentable stable category is dualizable, $\mathcal{C}(S)$ is locally rigid. This also follows by identifying it with the smashing ideal of p-torsion objects in Mod_S; compare [3, Example 1.38].

The inclusion $\mathcal{C}(S) \hookrightarrow \mathrm{Mod}_S$ preserves ω_1-filtered colimits. In fact, completeness is equivalent to $\mathrm{Hom}_S(S[1/p], M) = 0$, and $S[1/p]$ is a sequential colimit of copies of S, hence is ω_1-compact. Consequently completion preserves ω_1-compact objects, so the complete unit S is ω_1-compact. The tensor assertion follows from the compact generator and closure under countable colimits and retracts. These arguments apply equally to T.

The functor f carries K_S to K_T and hence preserves compact objects. Restriction commutes with completion, as follows from the formula $M_p^\wedge = \lim_n M/p^n$, so u preserves colimits in the complete categories. Each homotopy group of $u(K_T)$ is countably generated over $\pi_0 S$: it is an extension of modules of the forms $\pi_i T/p$ and $(\pi_{i-1}T)[p]$, which are finite over $\pi_0 T/p$. A finite type algebra is countably generated as a module over its coefficient ring.

A bounded below module over a bounded connective noetherian algebra with countably generated homotopy groups admits a countable cell presentation, by successively choosing countably many generators in increasing degrees. Noetherianity keeps the resulting kernels countably generated. Thus $u(K_T)$ is ω_1-compact in Mod_S, and also in $\mathcal{C}(S)$. Since the ω_1-compact objects of a compactly generated stable category are obtained from compact objects by countable colimits and retracts, u preserves them all. Finally, the projection formula is the usual associativity of tensor products followed by completion. □

2.2 The right adjoint to rigidification

For clarity, a presentable stable symmetric monoidal category is *locally rigid* if its underlying category is dualizable in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ and its multiplication has a continuous bimodule right adjoint. It is *rigid* if, in addition, its unit is compact. We use the following facts from [3, Propositions 1.9, 1.12, 1.32, 1.34 and Theorem 4.2].

If $\mathcal{C}$ is locally rigid with ω_1-compact unit, the canonical functor and its right adjoint are

$$q_{\mathcal{C}} : \mathcal{C}^{\mathrm{rig}} \rightleftarrows \mathcal{C} : \Lambda_{\mathcal{C}}. \tag{2.2}$$

The functor $\Lambda_{\mathcal{C}}$ is fully faithful, symmetric monoidal, exact, and preserves ω_1-filtered colimits. Its values on $\mathcal{C}^{\omega_1}$ generate $\mathcal{C}^{\mathrm{rig}}$ under colimits. There is an opposite-category realization

$$\mathcal{C}^{\mathrm{rig}} \simeq \mathrm{Ind}\big((\mathcal{C}^{\omega_1})^{\mathrm{op}}\big)^{\mathrm{rig}}, \qquad \Lambda_{\mathcal{C}}(P) \longmapsto (P^{\mathrm{op}})^\vee, \tag{2.3}$$

for $P \in \mathcal{C}^{\omega_1}$, where $X^\vee = \underline{\mathrm{Hom}}(X, \mathbf{1})$ in the indicated Ind-category. Moreover, a continuous symmetric monoidal functor between rigid categories has a continuous right adjoint. If $\mathcal{E}$ is rigid, an $\mathcal{E}$-module in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ is dualizable over $\mathcal{E}$ exactly when its underlying category is dualizable in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$.

For $\mathcal{C}(T)$ we write q_T and Λ_T. The next lemma makes the functoriality of (2.3) explicit.

Lemma 2.3 (Change of scalars)**.** *In the situation of Lemma 2.2, let $F = f^{\mathrm{rig}} : \mathcal{N}(S) \to \mathcal{N}(T)$ and let G be its right adjoint. The canonical mates are isomorphisms*

$$F\Lambda_S \simeq \Lambda_T f, \qquad G\Lambda_T \simeq \Lambda_S u. \tag{2.4}$$

Proof. The second isomorphism follows by taking right adjoints in the natural identity $q_T F = f q_S$. For the first, put $\mathcal{D}_S = \mathcal{C}(S)^{\omega_1}$ and $\mathcal{D}_T = \mathcal{C}(T)^{\omega_1}$. The adjunction $f \dashv u$ restricts to these small stable categories. Passing to opposites and then to Ind-categories gives an adjunction

$$\widetilde{u} = \mathrm{Ind}(u^{\mathrm{op}}) \dashv \widetilde{f} = \mathrm{Ind}(f^{\mathrm{op}}).$$

Both functors are continuous, $\widetilde{f}$ is symmetric monoidal, and the projection formula becomes

$$\widetilde{u}(Y \otimes \widetilde{f}X) \simeq \widetilde{u}(Y) \otimes X.$$

Consequently the canonical predual comparison is an isomorphism:

$$\begin{aligned} \mathrm{Hom}(Y, \widetilde{f}(X^\vee)) &\simeq \mathrm{Hom}(\widetilde{u}Y, X^\vee) \\ &\simeq \mathrm{Hom}(\widetilde{u}Y \otimes X, \mathbf{1}) \\ &\simeq \mathrm{Hom}(\widetilde{u}(Y \otimes \widetilde{f}X), \mathbf{1}) \\ &\simeq \mathrm{Hom}(Y \otimes \widetilde{f}X, \mathbf{1}). \end{aligned}$$

Thus $\widetilde{f}(X^\vee) \simeq (\widetilde{f}X)^\vee$.

Here is the compatibility with the canonical mate. In the realization $\mathcal{N}(S) \subset \mathrm{Ind}(\mathcal{D}_S)$, let n_S be the continuous right adjoint to the nuclear inclusion, and similarly for T. For $P \in \mathcal{D}_S$, one has $\Lambda_S(P) = n_S(P)$. The mate $F\Lambda_S(P) \to \Lambda_T(fP)$ is the mate of nuclearization for $\mathrm{Ind}(f)$. More explicitly, let

$$J_S : \mathrm{Ind}(\mathcal{D}_S) \longrightarrow \mathrm{Ind}(\mathcal{D}_S^{\mathrm{op}}), \qquad J_S(P) = (P^{\mathrm{op}})^\vee$$

be the continuous predual functor, and define J_T similarly. The calculation above gives a natural isomorphism $\widetilde{f} J_S \simeq J_T \,\mathrm{Ind}(f)$, first on compact objects and then on all objects by colimits. Proposition 1.34 of [3] identifies J_S applied to the nuclearization counit with an isomorphism, and likewise for T. Applying J_T to the nuclearization-mate square therefore identifies the mate with

$$\mathrm{Ind}(f^{\mathrm{op}})((P^{\mathrm{op}})^\vee) \longrightarrow ((fP)^{\mathrm{op}})^\vee;$$

this is the comparison just proved to be an isomorphism. Since J_T is an isomorphism on rigidifications [3, Proposition 1.32], the mate is an isomorphism. This proves the claim on ω_1-compact objects. Every object is an ω_1-filtered colimit of such objects, and both sides preserve those colimits. □

3 Full faithfulness and reduction modulo p

We use the following tensor-product criterion for rigid categories [4, Lemma 5.5].

Lemma 3.1 (Rigid Beck–Chevalley criterion)**.** *Given a commutative square of continuous symmetric monoidal functors between rigid categories*

$$\begin{array}{ccc} \mathcal{E}_0 & \xrightarrow{F_1} & \mathcal{E}_1 \\ {\scriptstyle F_2}\downarrow & & \downarrow{\scriptstyle G_1} \\ \mathcal{E}_2 & \xrightarrow[G_2]{} & \mathcal{E}_3, \end{array}$$

the induced functor $\mathcal{E}_1 \otimes_{\mathcal{E}_0} \mathcal{E}_2 \to \mathcal{E}_3$ is fully faithful if and only if the canonical mate

$$F_2 F_1^R \longrightarrow G_2^R G_1$$

is an isomorphism. Here the superscript R denotes a right adjoint.

Proof. For a continuous symmetric monoidal functor between rigid categories, the right adjoint is continuous and satisfies the projection formula. Full faithfulness is therefore equivalent to the adjunction unit being an isomorphism on the tensor unit. Under relative self-duality,

$$\mathcal{E}_1 \otimes_{\mathcal{E}_0} \mathcal{E}_2 \simeq \mathrm{Fun}^{\mathrm{L}}_{\mathcal{E}_0}(\mathcal{E}_1, \mathcal{E}_2),$$

that unit map is the displayed Beck–Chevalley mate. □

Proposition 3.2. *For A, B as in Lemma 2.1, put $C = A \mathbin{\widehat{\otimes}}_R B$. The canonical functor*

$$\Phi_{A,B} : \mathcal{N}(A) \otimes_{\mathcal{E}} \mathcal{N}(B) \longrightarrow \mathcal{N}(C)$$

is fully faithful.

Proof. Apply Lemma 3.1 to the square of rigidified pullback functors associated with R, A, B, C. Use its notation F_1, F_2, G_1, G_2, with $\mathcal{E}_1 = \mathcal{N}(A)$ and $\mathcal{E}_2 = \mathcal{N}(B)$. The two functors in the Beck–Chevalley mate preserve colimits. It is therefore enough to test that mate on $\Lambda_A(M)$ for $M \in \mathcal{C}(A)^{\omega_1}$.

Write $u_A : \mathcal{C}(A) \to \mathcal{C}(R)$ and $u_B : \mathcal{C}(C) \to \mathcal{C}(B)$ for restriction. By Lemma 2.3, the mate on this object identifies with Λ_B applied to

$$B \mathbin{\widehat{\otimes}}_R u_A(M) \longrightarrow u_B(C \mathbin{\widehat{\otimes}}_A M).$$

This is the canonical completed base-change isomorphism, since $C = A \mathbin{\widehat{\otimes}}_R B$. The identification is an identification of mates, by naturality and pasting of the mates in (2.4). Lemma 3.1 gives full faithfulness. □

Lemma 3.3. *Let T be a bounded noetherian p-complete commutative R-algebra. For every $Z \in \mathcal{N}(T)$ one has*

$$Z/p := \mathrm{cofib}(p : Z \to Z) \;\in\; \mathrm{Loc}(\mathbf{1}/p).$$

Consequently, if a full stable localizing subcategory $\mathcal{I} \subseteq \mathcal{N}(T)$ contains the unit, then multiplication by p is an isomorphism on every object of $\mathcal{N}(T)/\mathcal{I}$.

Proof. Let $T_1 = T \otimes_R \mathbb{F}_p$. All T_1-modules are p-complete, and Mod_{T_1} is already rigid. Reduction induces a continuous symmetric monoidal functor

$$H : \mathcal{N}(T) \longrightarrow \mathrm{Mod}_{T_1}.$$

Its right adjoint U is continuous. Taking right adjoints of the square with the rigidification counits gives

$$U(N) = \Lambda_T(\mathrm{res}\, N).$$

The category Mod_{T_1} is generated by T_1, and

$$U(T_1) = \Lambda_T(T \otimes_R \mathbb{F}_p) = \mathbf{1}/p.$$

It follows that $\Lambda_T(M \otimes_R \mathbb{F}_p)$ belongs to $\mathrm{Loc}(\mathbf{1}/p)$ for every complete T-module M. By exactness of Λ_T, this object is $\Lambda_T(M)/p$. The objects $\Lambda_T(M)$ generate $\mathcal{N}(T)$ under colimits, and $(-)/p$ preserves colimits. This proves the first assertion. In the quotient by $\mathcal{I}$, the cofiber of multiplication by p is therefore always zero, which proves the second assertion. □

4 Coherent generation of the generic diagonal

The essential-surjectivity argument uses coherent modules on ordinary affinoid algebras. We give the generation proof, since completed tensor products require an analytic input.

Lemma 4.1. *Let H be a noetherian ring of finite dimension and let $h \in H$. If $H[1/h]$ is regular, then*

$$\mathrm{Coh}^b(H) = \mathrm{thick}(H, \mathrm{Coh}^b_{V(h)}(H)),$$

where $\mathrm{Coh}^b_{V(h)}(H)$ consists of the bounded coherent modules whose localization at h vanishes.

Proof. Put $\mathcal{H} = \mathrm{thick}(H, \mathrm{Coh}^b_{V(h)}(H))$. First suppose that a finite discrete H-module M becomes projective after inverting h. Choose a finite projective complement W such that $M[1/h] \oplus W \simeq H[1/h]^n$. A finite presentation of W can be cleared of denominators, giving a finite H-module L with $L[1/h] \simeq W$. Clearing the images of a basis now gives a map

$$H^n \longrightarrow M \oplus L$$

which is an isomorphism after inverting h. Its cofiber belongs to $\mathrm{Coh}^b_{V(h)}(H)$, so $M \oplus L$, and hence M, belongs to $\mathcal{H}$.

For a general finite H-module, take successive finite free surjections and their kernels. Since $H[1/h]$ is regular of finite dimension, it has finite global dimension [9, Tag 00OE]. After finitely many steps the localized kernel is projective. The preceding argument, followed by the finite sequence of cofiber sequences, gives the assertion for the original module. Finite Postnikov filtrations give it for every bounded coherent module. □

Proposition 4.2 (Affinoid exterior generation)**.** *Let a, b be affinoid K-algebras, and put $c = a \mathbin{\widehat{\otimes}}_K b$, with ordinary completed tensor product. Then*

$$\mathrm{Coh}^b(c) = \mathrm{thick}\left\{P \mathbin{\widehat{\otimes}}_K Q \;\middle|\; P \text{ finite discrete over } a,\ Q \text{ finite discrete over } b\right\}. \tag{4.1}$$

Proof. Completed exterior products are exact on finite modules, as recalled in Lemma 2.1. In particular they commute with quotients by ideals in either factor. We argue by induction on $\dim a + \dim b$, with the zero ring causing no difficulty.

Finite nilpotent filtrations reduce the assertion to reduced a and b. More precisely, the images of their nilradicals generate a nilpotent ideal in c, and its quotient is $a_{\mathrm{red}} \mathbin{\widehat{\otimes}}_K b_{\mathrm{red}}$. Filtering a finite c-module by this ideal reduces to modules over that quotient. An exterior product for the reduced quotients is also an exterior product for a and b.

Suppose therefore that both rings are reduced. Over the perfect field K, a reduced affinoid has a dense smooth locus, whose complement is a closed analytic subset [2, §3.3, especially Lemma 3.3.1 and the paragraph following its proof]. By prime avoidance, choose $f \in a$ and $g \in b$ avoiding all minimal primes such that their nonvanishing loci are smooth. Set $h = (f \otimes 1)(1 \otimes g) \in c$.

The ordinary ring $c[1/h]$ is regular. To see this, recall that an affinoid algebra is Jacobson. Maximal ideals of $c[1/h]$ correspond to maximal ideals of c avoiding h [9, Tag 00G6]. Their rigid points lie in the product of the two smooth loci, which is smooth over K. At such a point, the algebraic local ring of c and the analytic local ring have the same completion [2, pp. 476–477]; hence the algebraic local ring is regular. This proves regularity of $c[1/h]$.

Lemma 4.1 reduces the proof to finite c-modules annihilated by a power of h, besides the unit c itself. If $h^n M = 0$, the exact sequence

$$0 \longrightarrow (f \otimes 1)^n M \longrightarrow M \longrightarrow M/(f \otimes 1)^n M \longrightarrow 0$$

has left term annihilated by $(1 \otimes g)^n$ and right term annihilated by $(f \otimes 1)^n$. Filtration by powers of these elements reduces to finite modules over

$$c/(f \otimes 1) \simeq (a/f) \mathbin{\widehat{\otimes}}_K b, \qquad c/(1 \otimes g) \simeq a \mathbin{\widehat{\otimes}}_K (b/g).$$

The dimensions decrease, so induction applies. The unit is itself an exterior product. Finite Postnikov filtrations complete the proof. □

Remark 4.3. The induction in Proposition 4.2 is the affinoid analogue of the coherent exterior-generation argument for finite type schemes over a field; compare [5, Lemma B.1.6]. Its use here is over the perfect field K.

Lemma 4.4 (Coherent localization)**.** *Let D be a bounded connective noetherian commutative R-algebra. For $P, Q \in \mathrm{Coh}^b(D)$, the canonical map*

$$\mathrm{Hom}_D(P, Q)[1/p] \xrightarrow{\sim} \mathrm{Hom}_{D[1/p]}(P[1/p], Q[1/p]) \tag{4.2}$$

is an isomorphism. Every bounded coherent $D[1/p]$-module is obtained from localizations of bounded coherent D-modules. Equivalently, localizing the mapping spectra of $\mathrm{Coh}^b(D)$ at p gives $\mathrm{Coh}^b(D[1/p])$.

Proof. A bounded coherent D-module is almost perfect and admits perfect approximations with arbitrarily connective cofiber [7, Proposition 7.2.4.17 and its proof]. For any prescribed homotopy group of $\mathrm{Hom}_D(P, Q)$, choose a perfect approximation to P whose cofiber is sufficiently connective relative to the upper homotopy bound of Q. This approximation calculates that homotopy group both before and after localization. Mapping out of a perfect object commutes with localization, proving (4.2) in every degree.

A finite $\pi_0 D[1/p]$-module admits a finite $\pi_0 D$-module lattice, by clearing a finite presentation. Regard that lattice as a D-module. A bounded coherent object over $D[1/p]$ has a finite Postnikov filtration with factors of this kind. By (4.2), its extension maps lift after multiplication by powers of p. Rescaling an extension map by such a power does not change its cofiber after inverting p. Induction on the filtration therefore lifts the object. □

Proposition 4.5 (An integral factorization of the diagonal)**.** *Let A be an ordinary topologically finite type, p-complete R-algebra and put $D = A \mathbin{\widehat{\otimes}}_R A$. Regard A as a D-module through multiplication. There exist finite discrete A-modules $P_1, \dots, P_r, Q_1, \dots, Q_r$, an object*

$$E \in \mathrm{thick}\{P_i \mathbin{\widehat{\otimes}}_R Q_i : 1 \le i \le r\} \subseteq \mathrm{Coh}^b(D),$$

and maps $A \xrightarrow{i} E \xrightarrow{\rho} A$ whose composite is $p^N \,\mathrm{id}_A$ for some integer $N \ge 0$.

Proof. The D-module A is bounded coherent. By Lemma 2.1, $D[1/p] = a \mathbin{\widehat{\otimes}}_K a$ for the affinoid algebra $a = A[1/p]$. Proposition 4.2 places its diagonal a in the thick subcategory generated by finite coherent exterior products. Each finite a-module admits a finite A-module lattice. Thus the generators involved can be written as $(P_i \mathbin{\widehat{\otimes}}_R Q_i)[1/p]$.

Choose an object E_K obtained by finitely many shifts and cofibers from these exterior products such that a is a retract of E_K. Lemma 4.4 permits lifting this finite construction to an object E over D, clearing denominators at each map. All the integral exterior products are bounded coherent by Lemma 2.1. Lift the retraction maps after multiplying them by powers of p, to maps $i_0 : A \to E$ and $r_0 : E \to A$. For some $a_0, b_0 \ge 0$, their localized composite equals $p^{a_0+b_0}\,\mathrm{id}$. Equation (4.2) then gives $c_0 \ge 0$ with

$$p^{c_0}(r_0 i_0 - p^{a_0+b_0}\,\mathrm{id}_A) = 0 \quad \text{in } \pi_0 \,\mathrm{End}_D(A).$$

Take $i = i_0$, $\rho = p^{c_0} r_0$ and $N = a_0 + b_0 + c_0$. If $a = 0$, the same argument starts with $E_K = 0$ and gives the assertion with $E = 0$. □

5 The affine tensor product formula

Theorem 5.1. *Let A, B be ordinary topologically finite type R-algebras complete for the p-adic topology, and put $C = A \mathbin{\widehat{\otimes}}_R B$. Then*

$$\Phi_{A,B} : \mathcal{N}(A) \otimes_{\mathcal{E}} \mathcal{N}(B) \xrightarrow{\sim} \mathcal{N}(C)$$

is an isomorphism of presentable stable symmetric monoidal categories.

Proof. Proposition 3.2 proves full faithfulness. Let $\mathcal{I} \subseteq \mathcal{N}(C)$ be the essential image. It is a full stable localizing subcategory and contains the unit. Form the presentable stable quotient $\mathcal{Q} = \mathcal{N}(C)/\mathcal{I}$. By Lemma 3.3, multiplication by p is an isomorphism on every object of $\mathcal{Q}$.

Fix any $M \in \mathcal{C}(C)$. Apply Proposition 4.5 to A, and distinguish the left and right copies of A in $D = A \mathbin{\widehat{\otimes}}_R A$. The exact functor

$$\mathrm{Coh}^b(D) \longrightarrow \mathcal{C}(C), \qquad L \longmapsto L \mathbin{\widehat{\otimes}}_{A,\mathrm{right}} M$$

uses the original A-action on M for the tensor product. The remaining left A-action and the B-action on M give the resulting C-action. It sends the diagonal A to M. On an exterior product it gives

$$(P_i \mathbin{\widehat{\otimes}}_R Q_i) \mathbin{\widehat{\otimes}}_{A,\mathrm{right}} M \simeq P_i \mathbin{\widehat{\otimes}}_R (Q_i \mathbin{\widehat{\otimes}}_A M), \tag{5.1}$$

where $Q_i \mathbin{\widehat{\otimes}}_A M$ is a complete B-module. Hence $p^N \mathrm{id}_M$ factors through an object in the thick subcategory generated by the right sides of (5.1).

The functor Λ_C is exact. For a complete A-module P and a complete B-module L, its monoidality and Lemma 2.3 give

$$\Lambda_C(P \mathbin{\widehat{\otimes}}_R L) \simeq G_1\Lambda_A(P) \otimes G_2\Lambda_B(L), \tag{5.2}$$

where $G_1 : \mathcal{N}(A) \to \mathcal{N}(C)$ and $G_2 : \mathcal{N}(B) \to \mathcal{N}(C)$ are the two pullbacks. The right side lies in $\mathcal{I}$. It follows that $p^N \mathrm{id}_{\Lambda_C(M)}$ factors through an object of $\mathcal{I}$.

In $\mathcal{Q}$ this endomorphism is zero, while multiplication by p^N is invertible. Thus the image of $\Lambda_C(M)$ in $\mathcal{Q}$ is zero. The objects $\Lambda_C(M)$ generate $\mathcal{N}(C)$ under colimits, so $\mathcal{Q} = 0$. Consequently $\mathcal{I} = \mathcal{N}(C)$, proving essential surjectivity. □

6 Zariski descent and formal schemes

We need descent in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$, so that it can be combined with relative tensor products. The following proof also covers the non-discrete formal schemes in (1.1).

Lemma 6.1. *Let $\mathfrak{Z}$ be a quasi-compact, quasi-separated p-adic formal scheme with connective structure sheaf, locally of the form $\mathrm{Spf}\, T$ with T bounded noetherian. Then $\mathrm{QC}(\mathfrak{Z})$ is locally rigid and its unit is ω_1-compact. For a quasi-compact open immersion $j : U \hookrightarrow \mathfrak{Z}$, restriction is a smashing localization*

$$j^* : \mathrm{QC}(\mathfrak{Z}) \longrightarrow \mathrm{QC}(U), \qquad j_*j^*(-) \simeq j_*\mathcal{O}_U \otimes (-),$$

and $j_*\mathcal{O}_U$ is ω_1-compact.

Proof. Let $i : \mathfrak{Z}_1 \to \mathfrak{Z}$ be the derived zero locus of p. The space $\mathfrak{Z}_1$ is a quasi-compact, quasi-separated spectral scheme. Its quasi-coherent category is compactly generated by perfect objects [6, Proposition 9.6.1.1]. The functor

$$i_* : \mathrm{QC}(\mathfrak{Z}_1) \longrightarrow \mathrm{QC}(\mathfrak{Z})$$

preserves colimits: this can be checked locally, where colimits of $\mathfrak{Z}_1$-modules remain p-complete. It has a continuous right adjoint $i^!$, given by $i^!M \simeq i^*M[-1]$. This formula follows locally from the self-duality, up to shift, of the perfect module $\mathrm{cofib}(p : \mathcal{O}_{\mathfrak{Z}} \to \mathcal{O}_{\mathfrak{Z}})$. It is conservative because reduction modulo p detects zero complete modules. The images under i_* of compact generators therefore compactly generate $\mathrm{QC}(\mathfrak{Z})$. Those images are locally perfect over $\mathcal{O}_{\mathfrak{Z}}$, since $i_*\mathcal{O}_{\mathfrak{Z}_1}$ is perfect, so they are dualizable. As in Lemma 2.2, this proves local rigidity.

The same argument for the derived zero locus of p^n shows that

$$K_n = \mathrm{cofib}(p^n : \mathcal{O}_{\mathfrak{Z}} \to \mathcal{O}_{\mathfrak{Z}})$$

is compact. Give these objects the transition maps induced by the identity on the source and multiplication by p on the target. In the complete category the telescope of multiplication by p on the unit is zero. Therefore

$$\mathcal{O}_{\mathfrak{Z}} \simeq \operatorname*{colim}_n K_n[-1],$$

which proves ω_1-compactness of the unit.

The statements about j may be checked on affine opens. A quasi-compact open in $\operatorname{Spf} T$ has a finite cover by principal formal opens. The pushforward of the unit is calculated by the finite diagram of nonempty intersections, whose terms are

$$T[1/f_J]^{\wedge}_p,$$

where f_J ranges over the products of the finitely many defining elements. Each such object is a sequential colimit of the unit in $\mathcal{C}(T)$, and is therefore ω_1-compact. The same finite Cech calculation proves continuity of j_* and its projection formula. Restriction has fully faithful right adjoint j_*, hence is the asserted smashing localization.

Finally, ω_1-compactness is local for finite affine covers in this setting. Mapping spectra for a union of two open subsets are the pullback of the mapping spectra on the two opens and their intersection. A quasi-compact open inside an affine has a finite principal cover with affine intersections. Induction on a finite affine cover, using these finite Cech formulas, proves the locality assertion. In particular it applies to $j_*\mathcal{O}_U$. □

Proposition 6.2 (Descent after rigidification)**.** *On the formal schemes of Lemma 6.1, the assignment $\mathfrak{Z} \mapsto \mathcal{N}(\mathfrak{Z})$ satisfies Zariski descent in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ and in $\mathrm{Mod}_{\mathcal{E}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. Restrictions to quasi-compact open subsets are smashing localizations.*

Proof. We apply the categorical smashing-localization results of [8, Corollary A.4 and Proposition A.5]. They assert the following. If $\mathcal{C}$ is locally rigid with ω_1-compact unit and a smashing localization is represented by an ω_1-compact idempotent e, its rigidification is the smashing localization represented by $\Lambda_{\mathcal{C}}(e)$. Moreover, a pair of such localizations with zero intersection of kernels gives a Cartesian square in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ after rigidification.

Lemma 6.1 verifies all these hypotheses for open restrictions. For a two-open cover the kernels have zero intersection, and the common localization is restriction to the intersection. Thus the usual two-open descent square remains Cartesian in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. Finite covers follow by induction, and quasi-compactness reduces arbitrary Zariski covers to finite ones. This proves descent. This application uses the categorical results of the cited appendix, which apply to the connective structure sheaves here without a discreteness assumption. The classical case is also [8, Proposition 2.8].

All functors carry their natural $\mathcal{E}$-linear structures. The forgetful functor $\mathrm{Mod}_{\mathcal{E}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}) \to \mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$ creates limits, so the same descent statement holds in $\mathrm{Mod}_{\mathcal{E}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$. □

Proof of Theorem 1.2. Every $\mathcal{N}(\mathfrak{W})$ occurring here is rigid and is therefore dualizable in $\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}}$. Since $\mathcal{E}$ is rigid, it is also dualizable as an $\mathcal{E}$-module [3, Proposition 1.12(iii)]. Tensoring by $\mathcal{N}(\mathfrak{W})$ over $\mathcal{E}$ consequently preserves limits in $\mathrm{Mod}_{\mathcal{E}}(\mathrm{Pr}^{\mathrm{L}}_{\mathrm{st}})$.

Take the limits over the affine open subsets $U \subseteq \mathfrak{X}$ and $V \subseteq \mathfrak{Y}$, with restriction as transition functor. By Proposition 6.2, dualizability, and Theorem 5.1, there are natural isomorphisms

$$\begin{aligned}\mathcal{N}(\mathfrak{X}) \otimes_{\mathcal{E}} \mathcal{N}(\mathfrak{Y}) &\simeq \lim_U \lim_V (\mathcal{N}(U) \otimes_{\mathcal{E}} \mathcal{N}(V)) \\ &\simeq \lim_U \lim_V \mathcal{N}(U \times^{\mathrm{der}}_{\mathfrak{S}} V) \\ &\simeq \mathcal{N}(\mathfrak{X} \times^{\mathrm{der}}_{\mathfrak{S}} \mathfrak{Y}).\end{aligned}$$

For the last line, first use descent on $\mathfrak{Y}$ after inverse image to $U \times^{\mathrm{der}}_{\mathfrak{S}} \mathfrak{Y}$, and then descent on $\mathfrak{X}$ after inverse image to the full product. The affine isomorphisms are the canonical exterior-product maps, so this is the comparison in (1.2). The construction respects the symmetric monoidal structures.

If one factor is flat over R, the completed spectral tensor products on affine charts are discrete and agree with the ordinary completed tensor products. Thus the formal product is ordinary in that case. □

7 Completions, the polynomial example, and the flatness condition

Corollary 7.1. *Let X_0, Y_0 be finite type schemes over $\mathbb{Z}$, and let $\widehat{X}_0, \widehat{Y}_0$ be their formal completions along the closed fibers defined by p. Then*

$$\mathcal{N}(\widehat{X}_0) \otimes_{\mathcal{E}} \mathcal{N}(\widehat{Y}_0) \simeq \mathcal{N}\big(X_0 \widehat{\times^{\mathrm{der}}_{\mathrm{Spec}\,\mathbb{Z}}} Y_0\big).$$

If either completion is flat over R, the derived product can be replaced by the ordinary product.

Proof. For a finite type algebra $A_0 = \mathbb{Z}[t_1, \dots, t_d]/J$, exactness of noetherian completion gives

$$A_{0p}{}^{\wedge} \simeq R\langle t_1, \dots, t_d\rangle/(J).$$

These affine descriptions show that the completions satisfy Definition 1.1. They also identify their derived formal product with the completion of the indicated derived product: this identification can be checked after tensoring with $\mathbb{Z}/p^n$ for every n, where completion induces an isomorphism. Theorem 1.2 applies. □

Corollary 7.2. *For all $m, n \geq 0$, there is a canonical isomorphism*

$$\mathcal{N}(R\langle x_1, \dots, x_m\rangle) \otimes_{\mathcal{E}} \mathcal{N}(R\langle y_1, \dots, y_n\rangle) \simeq \mathcal{N}(R\langle x_1, \dots, x_m, y_1, \dots, y_n\rangle).$$

In particular,

$$\mathrm{QC}(\mathrm{Spf}\,\mathbb{Z}_p[x]^{\wedge}_p)^{\mathrm{rig}} \otimes_{\mathrm{QC}(\mathrm{Spf}\,\mathbb{Z}_p)^{\mathrm{rig}}} \mathrm{QC}(\mathrm{Spf}\,\mathbb{Z}_p[y]^{\wedge}_p)^{\mathrm{rig}} \simeq \mathrm{QC}(\mathrm{Spf}\,\mathbb{Z}_p[x, y]^{\wedge}_p)^{\mathrm{rig}}.$$

Proof. Restricted polynomial algebras over R are flat over R, and their completed tensor product is the restricted polynomial algebra in the combined variables. Apply Theorem 5.1. □

Example 7.3 (Failure for the ordinary product without flatness)**.** Take $\mathfrak{X} = \mathfrak{Y} = \operatorname{Spf} \mathbb{F}_p$. Since all $\mathbb{F}_p$-modules are complete, $\mathcal{N}(\mathbb{F}_p) = \operatorname{Mod}_{\mathbb{F}_p}$. Theorem 5.1 gives

$$\operatorname{Mod}_{\mathbb{F}_p} \otimes_{\mathcal{E}} \operatorname{Mod}_{\mathbb{F}_p} \simeq \operatorname{Mod}_{\mathbb{F}_p \otimes_R \mathbb{F}_p}.$$

The endomorphism spectrum of the tensor unit on the right is $\mathbb{F}_p \otimes_R \mathbb{F}_p$, with

$$\pi_i(\mathbb{F}_p \otimes_R \mathbb{F}_p) = \begin{cases} \mathbb{F}_p, & i = 0, 1, \\ 0, & i \neq 0, 1. \end{cases}$$

The ordinary formal fiber product is $\operatorname{Spf} \mathbb{F}_p$, whose rigidified quasi-coherent category has unit endomorphism spectrum $\mathbb{F}_p$, concentrated in degree zero. The two symmetric monoidal categories are therefore not isomorphic. Thus the derived product in Theorem 1.2 cannot be omitted in general.

References

[1] S. Bosch, U. Güntzer, and R. Remmert, *Non-Archimedean Analysis*, Grundlehren der mathematischen Wissenschaften **261**, Springer, Berlin, 1984. doi:10.1007/978-3-642-52229-1.

[2] B. Conrad, *Irreducible components of rigid spaces*, Ann. Inst. Fourier (Grenoble) **49** (1999), no. 2, 473–541. doi:10.5802/aif.1681.

[3] A. I. Efimov, *Localizing invariants of inverse limits*, 2025, version 2. arXiv:2502.04123v2.

[4] A. I. Efimov, *Rigidity of the category of localizing motives*, 2025, version 1. arXiv:2510.17010v1.

[5] A. I. Efimov and L. Positselski, *Coherent analogues of matrix factorizations and relative singularity categories*, Algebra Number Theory **9** (2015), no. 5, 1159–1292. doi:10.2140/ant.2015.9.1159.

[6] J. Lurie, *Spectral Algebraic Geometry*, February 2018. `https://www.math.ias.edu/~lurie/papers/SAG-rootfile.pdf`.

[7] J. Lurie, *Higher Algebra*, September 2017. `https://www.math.ias.edu/~lurie/papers/HA.pdf`.

[8] H. Matsukawa, *Reconstruction of Formal Schemes from Categories of Nuclear Modules*, 2026, version 1. arXiv:2607.10184v1.

[9] The Stacks Project Authors, *The Stacks Project*, 2026. Tags 0GBW, 00G6, and 00OE. `https://stacks.math.columbia.edu`.